

Geometry
Unit 13
Lesson 4 Practice

Name Answer Key
Date _____
Period _____

Use the following information for questions 1-4. Directed line segment $\overline{LT}$ is shown.



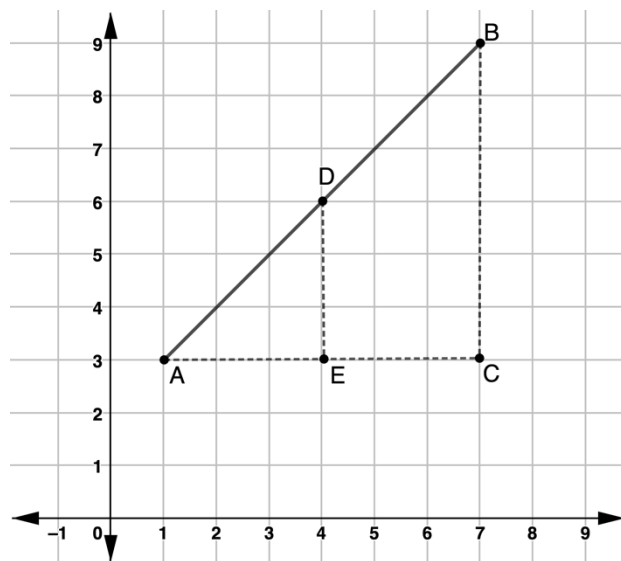
- How many equal parts is $\overline{LT}$ partitioned into? **6**
- Make a mark where $\overline{LT}$ is partitioned into a 1:3 ratio. Call this point K .
- Complete the statement.
Point K is $\frac{1}{4}$ of the distance from L to T .
- Complete the equation that can be used to find the distance of $\overline{LT}$.

$$LK = \frac{1}{4} LT$$

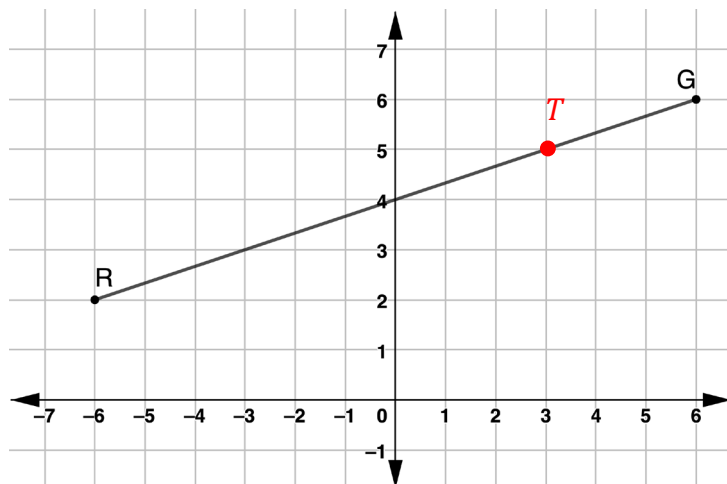
- Point D partitions directed line segment $\overline{AB}$ in a 3:3 ratio. Points $A(1,3)$, $E(4,3)$, $D(4,6)$, $B(7,9)$, and $C(7,3)$ are shown on the coordinate grid.

Use the distances to write the numeric ratio for each segment ratio. Then, choose the type of ratio the segment ratio represent.

Segment Ratio	Numeric Ratio	Type of Ratio
$AE:EC$	3:3	☐ part: part ☐ part: whole
$AE:AC$	3:6	☐ part: part ☐ part: whole
$DE:BC$	3:6	☐ part: part ☐ part: whole
$AD:DB$	3:3	☐ part: part ☐ part: whole



Use the following information for questions 6-8. Directed line segment RG is shown with endpoints at point $R(-6,2)$ and point $G(6,6)$.



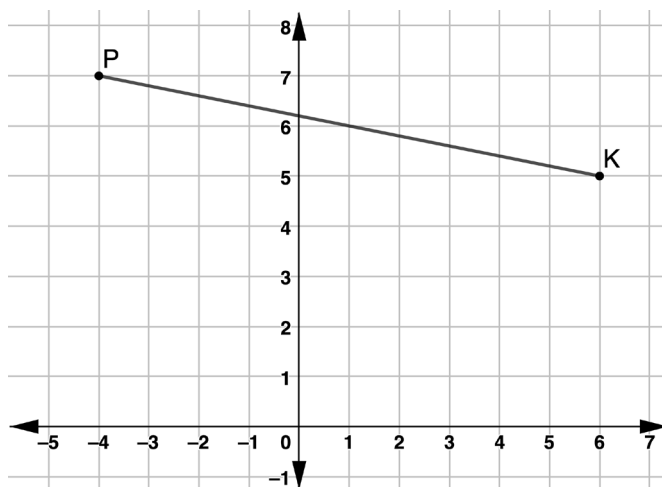
Directed line segment RG is partitioned $\frac{3}{4}$ of the distance from R to G at point T .

6. What is the horizontal distance of $\overline{RT}$? $12\left(\frac{3}{4}\right) = 9$

7. What is the vertical distance of $\overline{RT}$? $4\left(\frac{3}{4}\right) = 3$

8. What are the coordinates of point T ? $(3,5)$

Use the following information for questions 9-10. Directed line segment PK has endpoints at $P(-4,7)$ and $K(6,5)$.



Directed line segment PK is partitioned $\frac{2}{5}$ of the distance from P to K at point R .

9. What is the horizontal and vertical distance of $\overline{PR}$? $10\left(\frac{2}{5}\right) = 4$ $-2\left(\frac{2}{5}\right) = -\frac{4}{5}$

10. What are the coordinates of point R ? $\left(0, \frac{31}{5}\right)$